

TrES-3b

R-0168

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_220423 · created 2026-07-16T17:25:23+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459692.7951 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.015
Transit depth (Rp/Rs)^2	3.84 +/- 0.58 [%]
Orbital Inclination (inc)	84.47 +/- 1.17
Airmass coefficient 1 (a1)	0.986 +/- 0.021
Airmass coefficient 2 (a2)	0.0114 +/- 0.0063
Scatter in the residuals of the lightcurve fit is	2.14 %
Best Comparison Star	#3 - [522, 406]
Optimal Aperture	3.0
Optimal Annulus	11.57
Transit Duration (day)	0.0746 +/- 0.0059

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.015 vs 0.16309 (deviation 2.19σ)
Duration fit vs published	0.0746 d vs 0.0567 d (deviation 3.04σ)
Mid-time O–C	2.9 min
Depth SNR	13.1
Residual scatter	2.14 %

Light curve

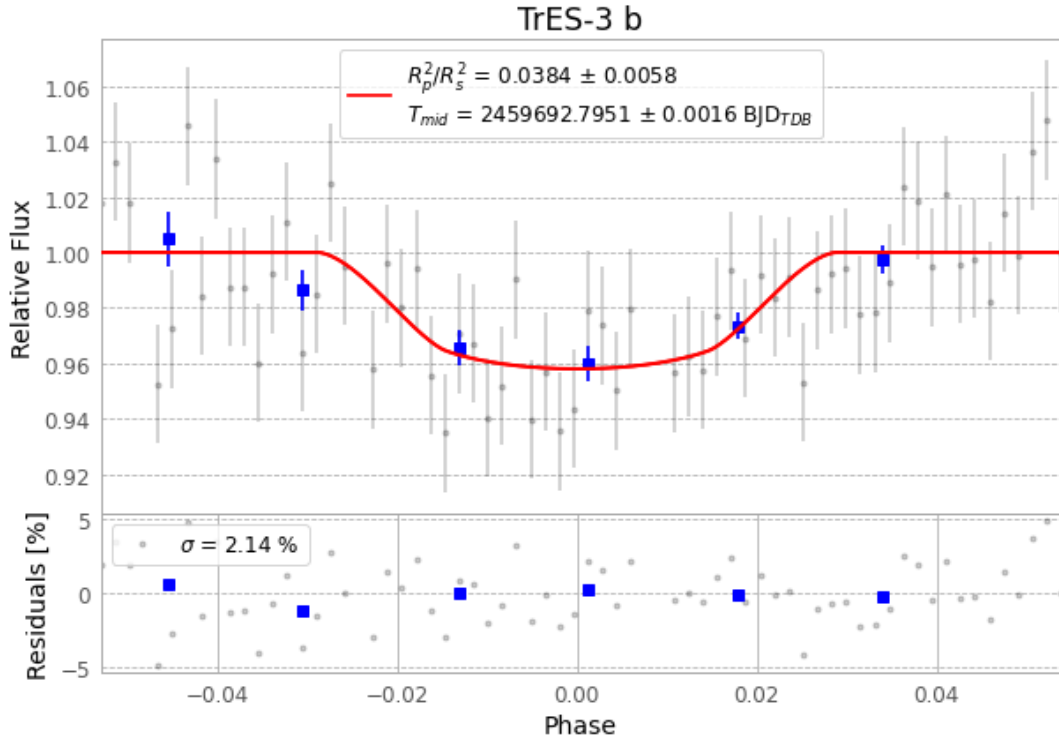


Figure 1 — FinalLightCurve_TrES-3 b_2022-04-22.png (SHA-256 8f06d46e2b036865...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [201, 317], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5c021ace7b475d36...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62be8a4

Data provenance

66 FITS frames (43 MB), TRES-3220423052114.FITS → TRES-3220423084213.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-220423.log (463a90b85ae7...), FinalLightCurve_TrES-3 b_2022-04-22.png (8f06d46e2b03...), FinalLightCurve_TrES-3 b_2022-04-22.csv (641426b04885...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 17:25Z.